Department of Electrical and Computer Engineering

North South University

Course: CSE331L – Microprocessor Interfacing and Embedded Systems Lab

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Section: 8

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Group Number: 3

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“Smart Parking System” Project Overview

Problem: Many small facilities lack affordable, automated control for a single parking bay—leading to confusion at the barrier, slow throughput, and safety risks near the gate.

What our prototype does: On power-up the LCD greets users, then cycles to show **temperature & humidity (DHT11)**. During approach, an **ultrasonic sensor (HC-SR04)** measures distance (and coarse speed) and drives **traffic-light LEDs**: **Green** (far), **Yellow** (mid), **Red** (close). At **close** distance the **servo barrier opens**. An **IR obstacle sensor** confirms the vehicle is fully inside the bay. The system then prompts for a **temporary 4-digit password** on a **4×4 keypad**; after entry it **locks** (servo closed) and shows **“Lock. Give PassWord”**. Re-entering the same code later **unlocks** and opens the barrier for exit; beam-break confirms departure and returns to idle.

Why it matters: Provides low-cost automation and clear human-machine interaction, improving safety and flow while demonstrating embedded topics (I²C LCD UI, timers, PWM servo, keypad scanning, debouncing, finite-state control).

Required sensors

Core Sensors and Software:

- HC-SR04 ultrasonic (distance & speed)
- IR obstacle sensor (beam-break)
- DHT11 (ambient RH for slippery hint)
- MCU: STM32F103C8 (Blue Pill), ST-LINK/V2, breadboard & jumpers.
- LCD 16×2 I²C (PCF8574 @ 0x27, 5 V) with PB6=SCL, PB7=SDA.
- 3× LEDs (G/Y/R) + resistors.
- SG90 Servo (5 V, separate supply ≥800 mA).
- Software: STM32CubeMX, Keil uVision 5, ST-Link Utility.

Functionalities

- LCD shows Dist (cm) and Speed (cm/s); page-flip to Temp/RH with *SLIPPERY?* hint when RH ≥ 80%.
- **Barrier logic:** auto-open on approach; beam-break confirms crossing; occupancy & dwell time tracked; auto-close with timeouts.
- LED traffic-light guidance: Green (free), Yellow (align), Red (stop/too close).
- **Stretch**: keypad PIN override; wet-floor detection via soil sensor on ramp; UART logs.

System Design

Hardware block. HC-SR04 (TRIG→PA1, ECHO→PA2 via divider), IR obstacle (PB11), DHT11 (PB10), 4×4 keypad (Rows: PA3–PA6, Cols: PB12–PB15), LCD 16×2 I²C (PB6/PB7 @ 0x27), LEDs G/Y/R (PB0/PB1/PB2), servo SG90 (PA8 PWM). All grounds common; servo on separate 5 V rail.

Data flow. Sensors → filtering (median for ultrasonic, debounce for IR/keypad) → finite-state logic → outputs (LCD, LEDs, servo).

Distance bands & actions.

- **Far (≥120 cm):** LED Green, gate closed, show `Dist: NNN cm`.
- **Mid (40–119 cm):** LED Yellow, guidance on LCD.
- **Close (<40 cm):** LED Red; open gate (servo). If IR beam breaks inside → transition to password stage.

Password & lock flow. After IR confirm, LCD prompts: *Set Temp Password*. Store 4 digits in RAM; close/lock gate; LCD shows **"Lock. Give PassWord"**. On correct re-entry later → open gate for exit; IR beam confirms leaving → clear temp password → idle.

Firmware architecture.

- **Tasks (10–20 Hz loop):** read sensors → compute distance band → update LEDs/LCD → run state machine → drive servo.
- **State machine:** `{IDLE → GATE_OPEN_ENTRY → SET_PW → LOCKED → GATE_OPEN_EXIT → IDLE}` with safety timeouts (8–12 s) on open states.

- **Timing:** SysTick (ms), TIM4 (1 MHz) for DHT11/HC-SR04 µs timing, TIM1 PWM (50 Hz) for servo. Keypad scanned row-by-row with pull-ups on columns.

Safety. Echo level shift; watchdog optional; servo brown-out avoidance via separate 5 V.

Design Rationale – Why We Chose This

- **Meets clear user need** (guided entry + secure lock) with minimal, low-cost sensors.
- **Strong pedagogy:** keypad scanning, I²C LCD UI, timing-critical ultrasonic, and a clean state machine for safety/timeouts.
- **Scalable:** thresholds/logic can be tuned; optional speed/buzzer/override PIN are easy extensions.
- **Safe interfacing:** explicit 5 V→3.3 V protection and separate servo supply.

How the proposal related to our course topics

The Parking systems are concrete applications of CSE331Lab concepts—exact same MCU, tools, peripherals, and design methods—showcasing the full stack from hardware wiring and driver code to real-time control and user interface.

“Smart Greenhouse & Water Tank Manager” Project

Overview

Problem: Manual watering often over/under-waters plants. Hobby greenhouses rarely track **plant comfort** or **tank level**, wasting water and harming growth.

What our prototype does. Measures **temperature & humidity (DHT11)**, **soil-moisture (ADC)**, and **tank level (HC-SR04)**. Computes **Heat Index**, **soil moisture %** (via DRY/WET calibration), and **tank fill %** (with known height). Shows results on **16×2 I²C LCD (0x27)**, indicates status with **G/Y/R LEDs**, and actuates an **SG90 servo** as a demo “valve” when soil is dry and water is available. Optional keypad sets thresholds; buzzer alerts on critical states.

Why it matters: Encourages water-efficient irrigation and healthier plants using inexpensive sensors, while exercising HAL drivers, filtering, and control logic.

Required sensors

Core Sensors and Software:

- DHT11 (Temp/RH)
- Soil-moisture sensor (analog) → ADC
- HC-SR04 ultrasonic for tank level (TRIG/echo)
- MCU: STM32F103C8 (Blue Pill), ST-LINK/V2, breadboard & jumpers.
- LCD 16×2 I²C (PCF8574 @ 0x27, 5 V) with PB6=SCL, PB7=SDA.
- 3× LEDs (G/Y/R) + resistors.
- SG90 Servo (5 V, separate supply ≥800 mA).

- Software: STM32CubeMX, Keil uVision 5, ST-Link Utility.

Functionalities

- LCD cycles: HI xx.x°C RH yy% / Soil zz% Tank ww%.
- LED status policy: Green (OK), Yellow (attention), Red (critical).
- Servo “valve” opens if **Soil < 40%** and **Tank ≥ 25%**.

Computed Metrics

- **Heat Index (°C)** from DHT11 temp/RH (Rothfusz regression).
- **Soil %** via calibrated ADC (DRY/WET).
- **Tank %** from distance with known tank height.
- **Stretch**: keypad thresholds, buzzer alarms, UART logging.

System Design

Hardware block. DHT11 (PB10), soil-moisture analog (PA0→ADC1_IN0), HC-SR04 for tank (TRIG→PA1, ECHO→PA2 via divider), LCD 16×2 I²C (PB6/PB7 @ 0x27), LEDs G/Y/R (PB0/PB1/PB2), servo SG90 (PA8 PWM). Common ground; servo on separate 5 V.

Data flow. Sensors → filtering (EMA/median) → metrics → thresholds → outputs.

Computed metrics.

- **Heat Index (°C):** from DHT11 Temp/RH (Rothfusz regression).
- **Soil moisture %:** map ADC between DRY/WET calibration points.
- **Tank fill %:** `(H − d)/H × 100` with known tank height *H* and distance *d*.

Actions & UI.

- LCD pages: `HI/Temp/RH` and `Soil%/Tank%`.
- LED policy: Green (OK), Yellow (attention), Red (critical).
- Servo "valve": open when soil < threshold and tank ≥ minimum; otherwise closed.

Firmware architecture.

- **Tasks (2–5 Hz):** read DHT/ADC/ultrasonic → compute metrics → update LED/LCD → servo control.
- **State bands:** `{OK, ATTENTION, CRITICAL}` per thresholds.
- **Calibration hooks:** store `ADC_DRY/ADC_WET`, tank height *H*.

Safety. Echo level shift; keep sensors dry; noise filtering; servo powered separately.

Design Rationale – Why We Chose This

- **Resource-efficient irrigation demo** using readily available sensors.
- **Meaningful analytics:** Heat Index, soil %, tank % → actionable decisions (servo valve).
- **Reusability:** drivers and filtering patterns carry over to many IoT/embedded builds.
- **Lab-friendly:** simple calibration steps; deterministic behavior via thresholds.

How the proposal related to our course topics

The Greenhouse systems are concrete applications of CSE331Lab concepts—exact same MCU, tools, peripherals, and design methods—showcasing the full stack from hardware wiring and driver code to real-time control and user interface.